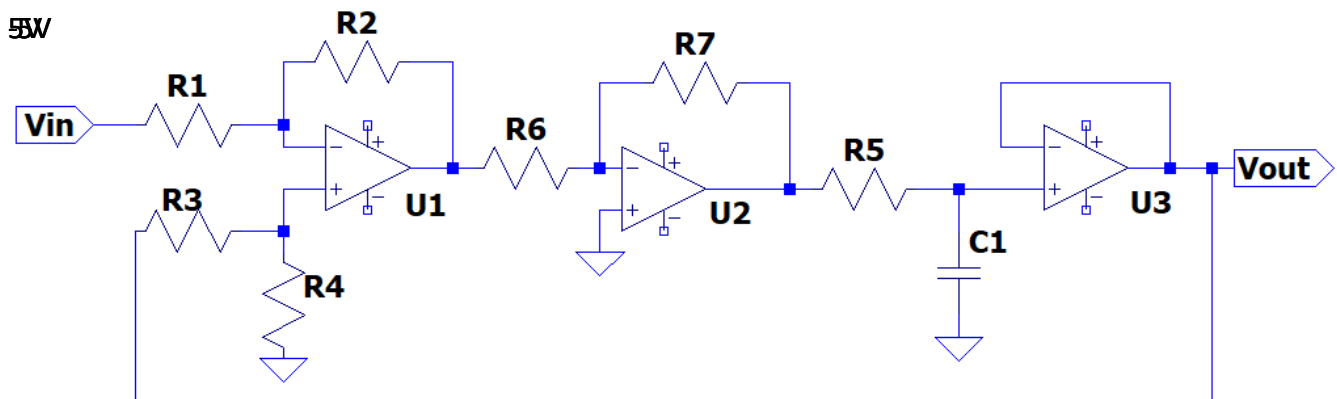


Experiment 1: RC circuit with unit feedback



	rise time (s)	delay time (s)	steady-state error (V)
Theoretical Result	0.2197	0.073733	0.5
Experiment Result	0.33	0.1	0.38
Simulation Result	0.2197	0.073733	0.5

1. Calculate the **closed loop transfer function**.

difference amplifier 為減法器的電路

Handwritten calculations for the closed loop transfer function:

Left side (Difference Amplifier):

$$V_{in} \cdot \left(\frac{R_2}{R_1} \cdot \left| \frac{R_7}{R_6} \right| \cdot \frac{1}{0.25s+1} \right) = V_{out}$$

open loop tf

$$= 1 \cdot 1 \cdot \frac{1}{0.25s+1} = \frac{1}{0.25s+1}$$

Block diagram showing a summing junction with input R and feedback path with transfer function $\frac{1}{0.25s+1}$ leading to output C.

close loop tf

$$= \frac{\frac{1}{0.25s+1}}{1 + \frac{1}{0.25s+1}} = \frac{1}{0.25s+2}$$

Right side (Plant):

plant

Block diagram showing input V_{in} through resistor R_5 to a node, then through capacitor C_1 to ground. The output is V_{out} . Notes: "buffer gain = 1", "Z → ∞", "no current in", "而前分 R_5, C_1 用分压", "因 i 同故 $V = iR$ ", "⇒ $V \propto R$ ".

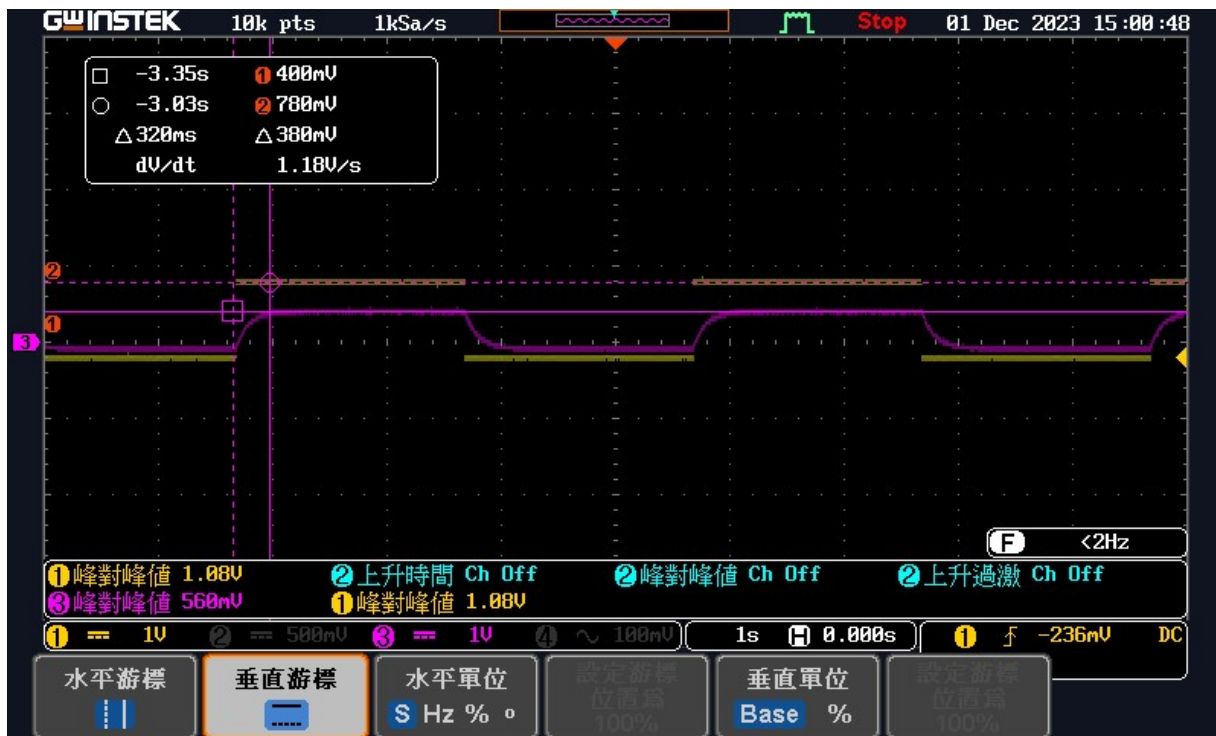
Transfer function:

$$G_p = \frac{\frac{1}{Cs}}{R_5 + \frac{1}{Cs}}$$

$$= \frac{1}{1 + R_5 C_1 s}$$

$$= \frac{1}{1 + 200k \cdot 1\mu s \cdot s}$$

$$= \frac{1}{0.25s+1}$$



Simulation

Closed loop transfer function from command window:

sys =

$$\frac{1}{0.2 s + 2}$$

The unit step response:

```
Gp =  
    1  
    ----  
    0.2 s + 1
```

Continuous-time transfer function.

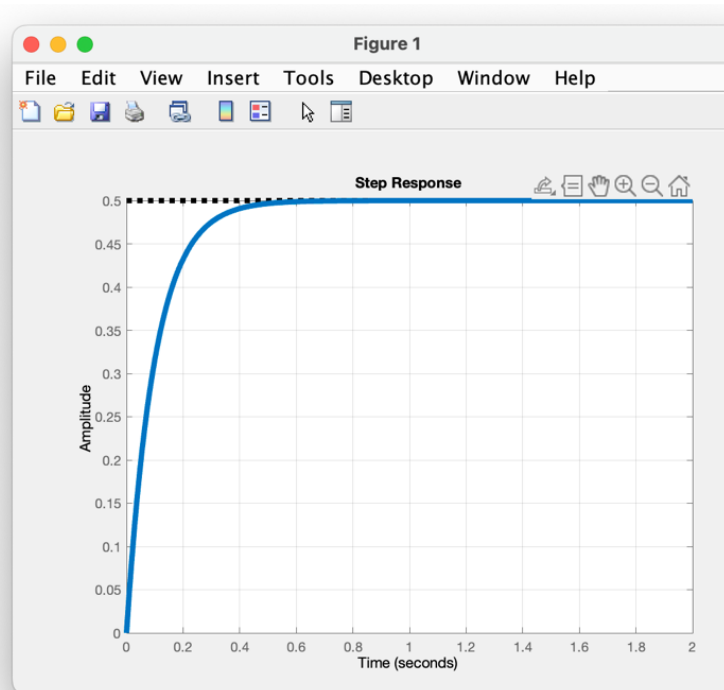
```
sys =  
    1  
    ----  
    0.2 s + 2
```

Continuous-time transfer function.

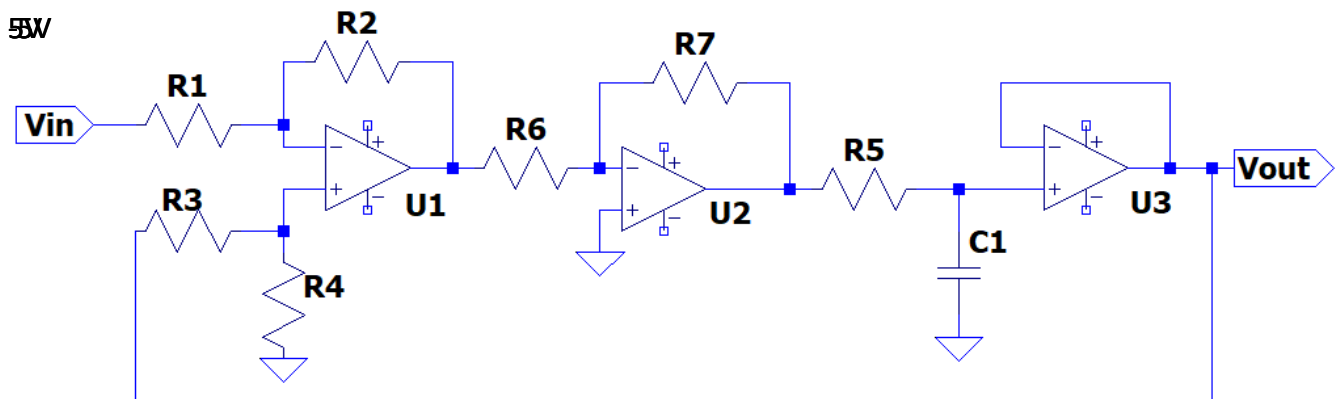
```
ans =  
struct with fields:  
    RiseTime: 0.2197  
    TransientTime: 0.3912  
    SettlingTime: 0.3912  
    SettlingMin: 0.4523  
    SettlingMax: 0.5000  
    Overshoot: 0  
    Undershoot: 0  
    Peak: 0.5000  
    PeakTime: 1.0546
```

```
steadyStateError =  
    0.5000
```

```
Delay Time (td): 0.073733 seconds
```



Experiment 2: RC circuit with P-controller and unit feedback

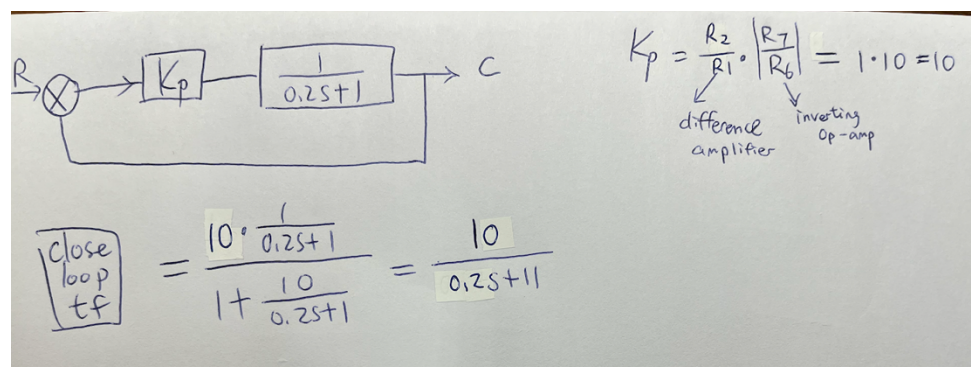


	rise time (s)	delay time (s)	steady-state error (V)
Theoretical Result	0.0399	0.0134	0.0909
Experiment Result	0.04	0.01	0.09
Simulation Result	0.0399	0.0134	0.0909

1. Calculate K_p and the closed loop transfer function.

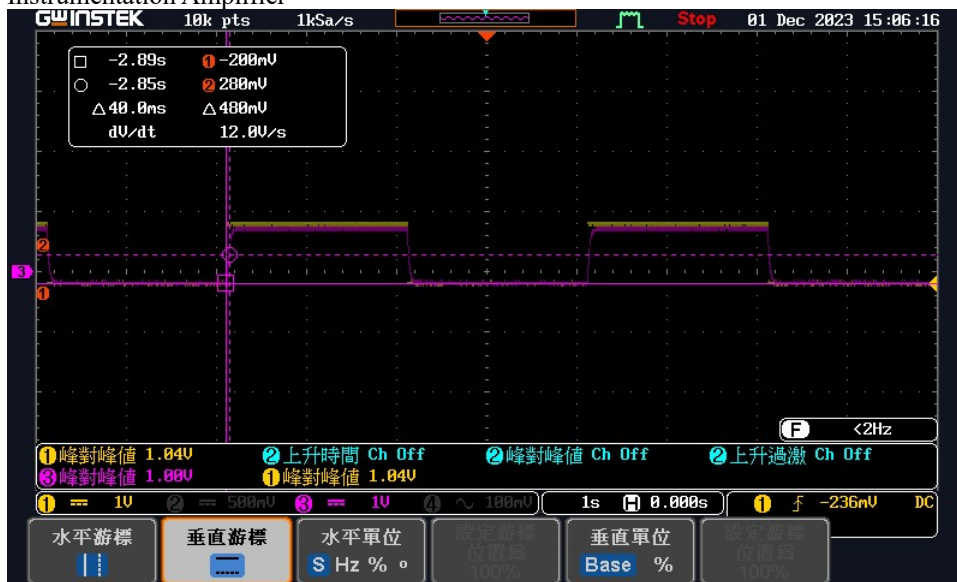
difference amplifier 為減法器的電路

$K_p = R_7/R_6 = 10$, plant 由上方小題可知



2.

V_{out} and V_{in} waveform (1 pic)



Simulation

Closed loop transfer function from command window:

sys =

10

0.2 s + 11

The unit step response:

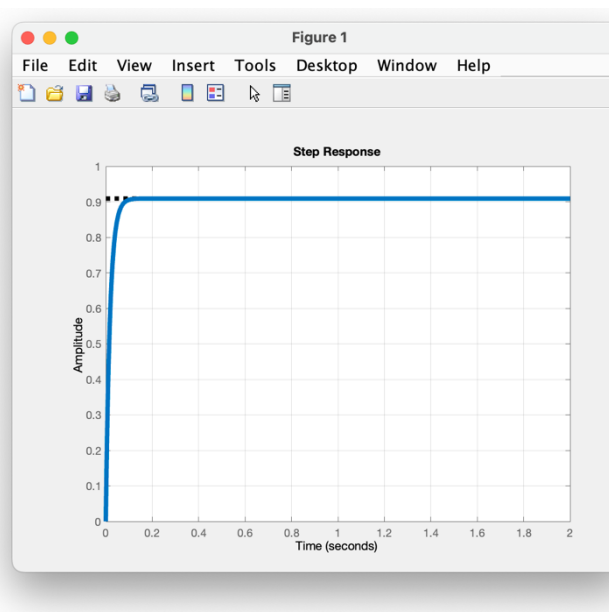
```
sys =
    10
    -----
    0.2 s + 11

Continuous-time transfer function.

ans =
    struct with fields:
        RiseTime: 0.0399
        TransientTime: 0.0711
        SettlingTime: 0.0711
        SettlingMin: 0.8223
        SettlingMax: 0.9091
        Overshoot: 0
        Undershoot: 0
        Peak: 0.9091
        PeakTime: 0.1917

steadyStateError =
    0.0909

Delay Time (td): 0.0134 seconds
fx >>
```



Discuss:

Try to explain the effect of the gain K_p on the overall system.

p-controller gain K_p 提高，則使 response time 變快（很快能到達穩態），因 p-control 會影響 pole 之位， K_p 提高，pole 的 real part 非常負，transient response 很快 decay

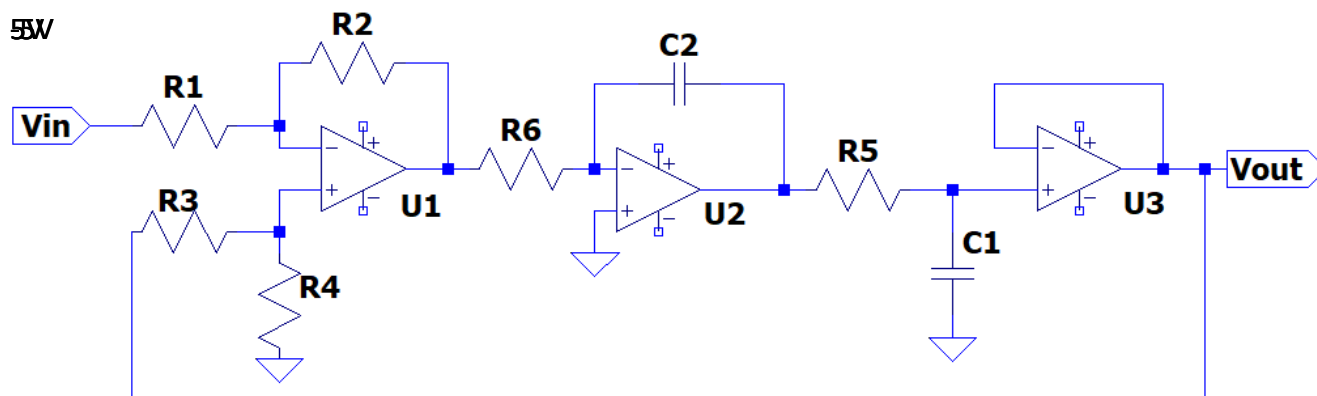
可以看到本題加入 P-control 後的 transient time 相比第一題單純 unit feedback 的 transient time 小了很多，且 p-controller gain K_p 提高，能 reduce steady state error，但仍無法完全消除 steady state error
assume step input and unit feedback, 且為一階系統，則其值等於 $1/(1+K_p)$,

故 p-controller gain K_p 提高則 steady state error 下降

,直觀來想，就是假設 error 為一個定值在特定的時間點，若 K_p 大，則輸出變快的大，if error 產生一些浮動，因此能更快的達到目標，但無法消除 steady state error, 因當 steady state error 很小時，就算 K_p 足夠大，但相乘仍然很小 input 改變量只有一點點，無法足夠趨近 desired output

可以看到本題加入 P-control 後的 steady state error 相比第一題單純 unit feedback 的 steady state error 小了很多。

Experiment 3: RC circuit with I-controller and unit feedback



	Maximum overshoot (%)	peak time (s)	rise time (s)	delay time (s)	settling time for 5% (s)	steady-state error (V)
Theoretical Result	70.2118	0.1405	0.0509	0.070225	1.563	1.9157 μ
Experiment Result	38.16	0.16	0.06	0.07	1.46	56m
Simulation Result	70.2118	0.1405	0.0509	0.070225	1.563	1.9157 μ

1. Calculate K_I and the closed loop transfer function.

$$K_I = 1/(10000C_2) = 100$$

difference amplifier 為減法器的電路,不用加入 K_I 上方誤寫

For 中 間

$$G_p = \frac{1}{0.25s+1}$$

$$\frac{0 - V_{in}}{10k} = \frac{V_{out}}{\frac{1}{Cs}}, \quad C = 1\mu$$

$$\Rightarrow V_{out} = -\frac{V_{in}}{Cs \cdot 10k}$$

$$\Rightarrow K_I = \frac{1}{10kC_2} = 100$$

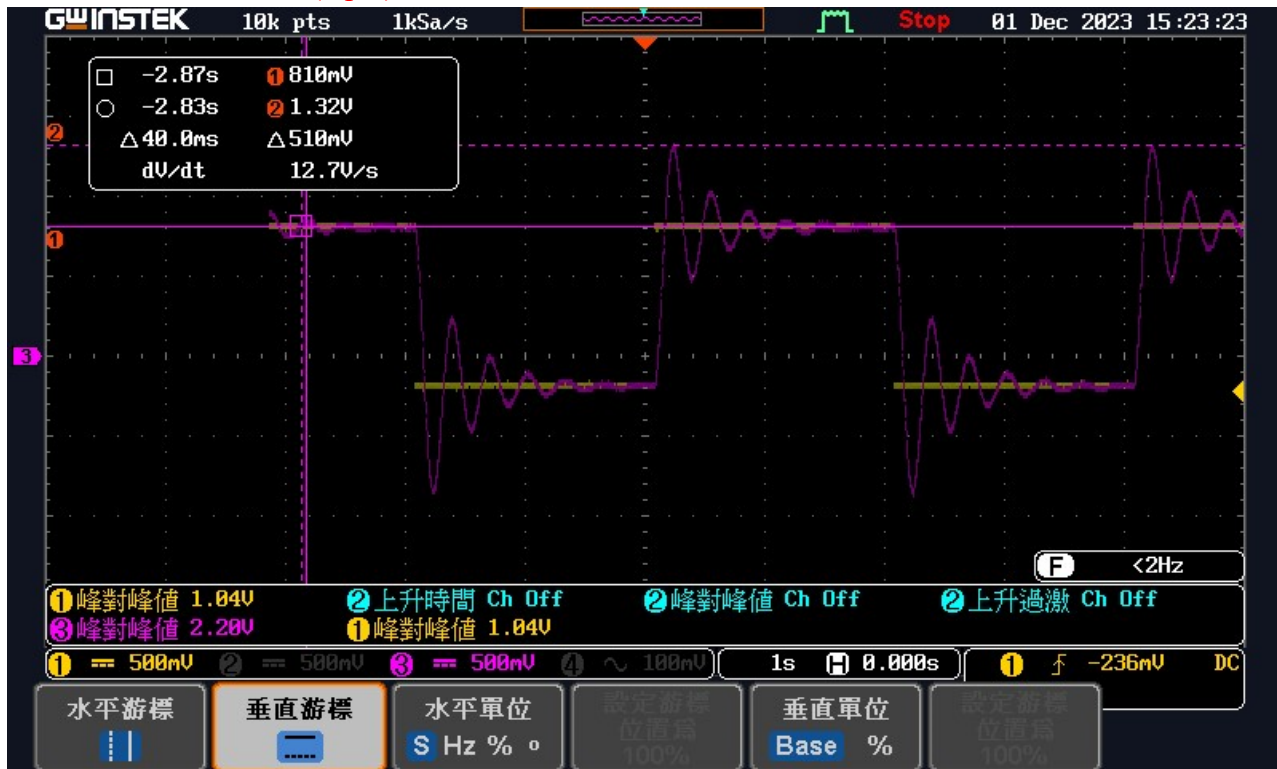
by difference amplifier $R_2 = R_1$ 故 Gain = 1

Close loop $\pm f$

$$\frac{C}{R} = \frac{K_I G_p}{1 + \frac{K_I G_p}{s}}$$

$$= \frac{\frac{100}{s} \cdot \frac{1}{0.25s+1}}{1 + \frac{100}{s} \cdot \frac{1}{0.25s+1}}$$

$$= \frac{1}{0.0025s^2 + 0.01s + 1}$$

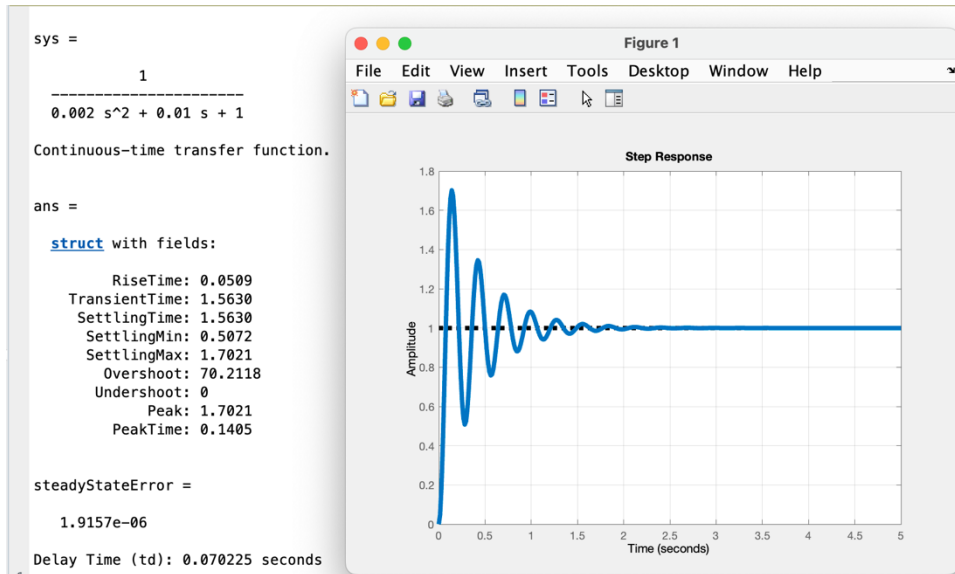
V_{out} and V_{in} waveform (1 pic):

2. Simulation

Closed loop transfer function from command window:

$$\text{sys} = \frac{1}{0.002 s^2 + 0.01 s + 1}$$

The unit step response:



Discuss:

Try to explain why the steady-state error is close to zero and the effect of the gain K_I on the overall system.

加入I-controller can reduce steady state error to

0 因為積分器能記錄過去的數值只要繼續有error存在，就算error再小也會持續的積起，使能持續的接近desired output，直到error等於零就不再加入值，另外steady state

error=0時，仍能有input（之前的累加），故不會產生 steady state

error=0就沒有input的情況，在圖中也可以看到加入I controller的穩態誤差相比第二小題加入P controller

小了許多，已經趨近於零了，若再加大觀察時間，能再持續的逼近零，但是也因為加入KI，overshoot 很大Maximum overshoot 大概為70%。KI很小會使系統響應變慢，但高KI

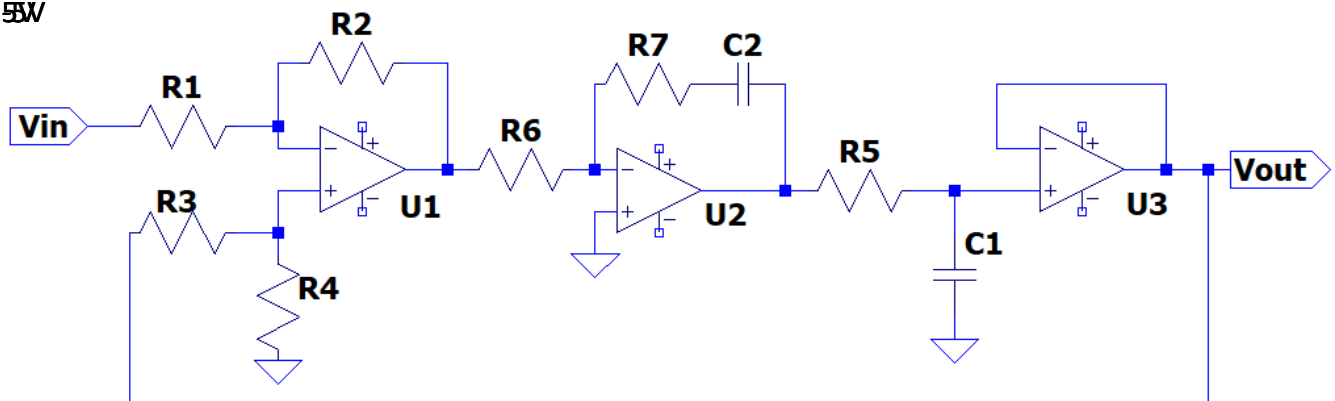
也可能在暫態的部分造成instability and

overshooting(由圖可見)，因為積分器紀錄是將過去和現在都用同等的比例積起，很早以前的數據已經不那麼管用了。I-

controller也可能造成系統不穩定(加入1/s可能使pole跑至右半面，指數係數變成正，而使數值趨近無窮，當時間趨近無窮。

Experiment 4: RC circuit with PI-controller and unit feedback

55V



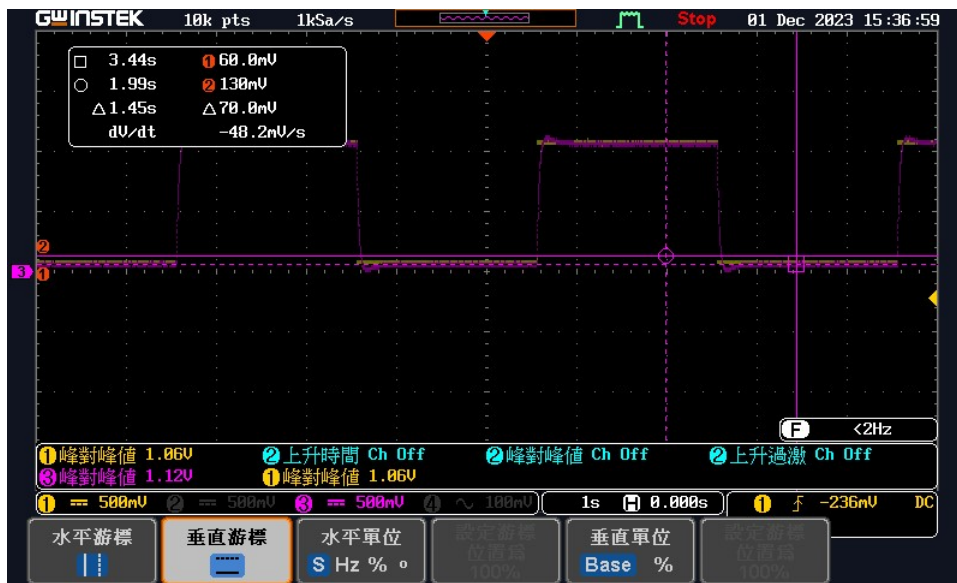
	Maximum overshoot (%)	peak time (s)	rise time (s)	delay time (s)	settling time for 5% (s)	steady-state error (V)
Theoretical Result	4.8838	0.0974	0.0358	0.01482	0.2007	$5.5511 \cdot 10^{-16}$
Experiment Result	7.8	0.11	0.04	0.02	0.2	51m
Simulation Result	4.8838	0.0974	0.0358	0.01482	0.2007	$5.5511 \cdot 10^{-16}$

Calculate K_p , K_I and the closed loop transfer function.

$K_p = \frac{R_7}{R_6} = 10$
 $K_I = \frac{1}{C_2 R_6}$
 $= \frac{1}{1\mu \cdot 10k}$
 $= 100$

plant $\frac{1}{0.2s+1}$

$\frac{C}{R} = \frac{(k_p + \frac{k_I}{s}) (\frac{1}{0.2s+1})}{1 + (k_p + \frac{k_I}{s}) (\frac{1}{0.2s+1})}$
 $= \frac{(10 + \frac{100}{s})}{0.2s+1 + k_p + \frac{k_I}{s}}$
 $= \frac{0.1s+1}{0.002s^2 + 0.1s+1}$

1. V_{out} and V_{in} waveform (1 pic):

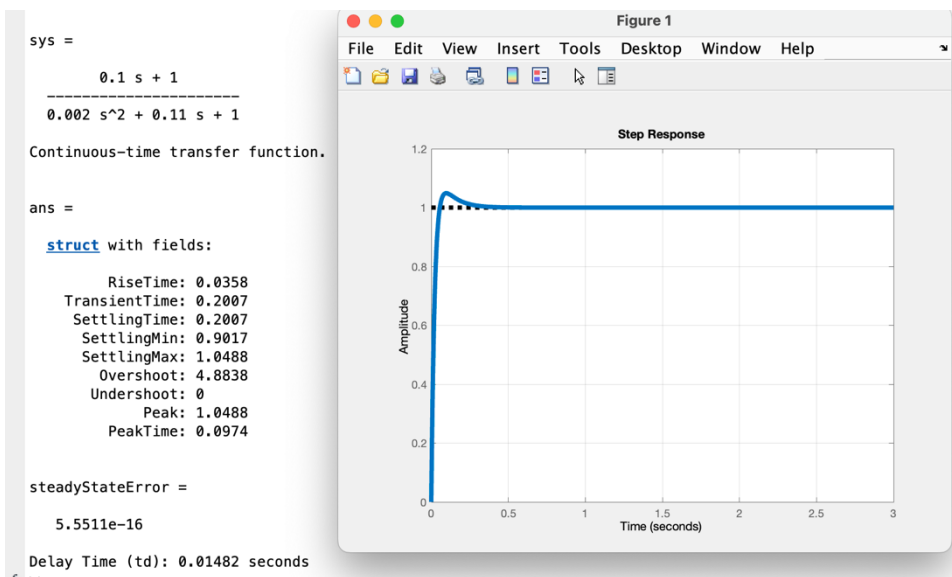
2. Simulation

Closed loop transfer function from command window:

sys =

$$\frac{0.1 s + 1}{0.002 s^2 + 0.11 s + 1}$$

The unit step response:



Discuss:

Try to explain the effect of PI-controller on the overall system and how to design K_p and K_I .

加入 KI 可以去除只有 K_p 會剩餘的 steady state error，增大 KI 可以增加系統反應速率，但過大會產生 overshoot，因此需加大 K_p ，改善 overshoot。可以由圖中看到加入 KI 後 overshoot 明顯相較上小題，只有 KI 的相比小了許多，Maximum overshoot 只有 4.88%，而穩態誤差也趨近於零。這題的 rise time 相比上題也小了一些，總之，可以看到如果想要消除穩態誤差，且減少 overshoot、加快反應速度，用一個 PI-controller，來改善系統的特性是一個不錯的方法。